

The Open Economy

Chapter 7

Econ 313 – Intermediate Macroeconomics

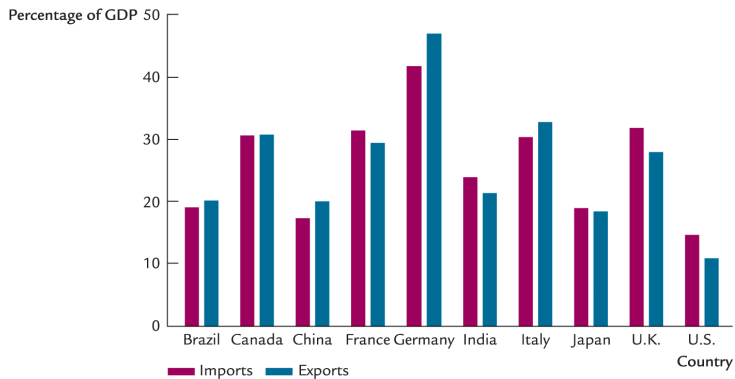
University of Tennessee, Knoxville

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In This Chapter, You Will Learn:

- ▶ Accounting identities for the open economy
- ▶ The small open-economy model
 - ▶ What makes it “small”
 - ▶ How the trade balance and exchange rate are determined
 - ▶ How policies affect trade balance and exchange rate

Imports and exports as a percentage of output, selected countries, 2021



In an open economy

- ▶ spending need not equal output
- ▶ saving need not equal investment



Port of Algeciras, Spain — one of Europe's busiest trade hubs.

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The national income identity in an open economy

$$Y = C + I + G + NX$$

or,

$$\underbrace{NX}_{\text{net exports}} = \underbrace{Y}_{\text{output}} - \underbrace{(C + I + G)}_{\text{domestic spending}}$$

Trade surpluses and deficits

$$NX = X - IM = Y - (C + I + G)$$

► **Trade surplus:**

output > spending and exports (X) > imports (IM)

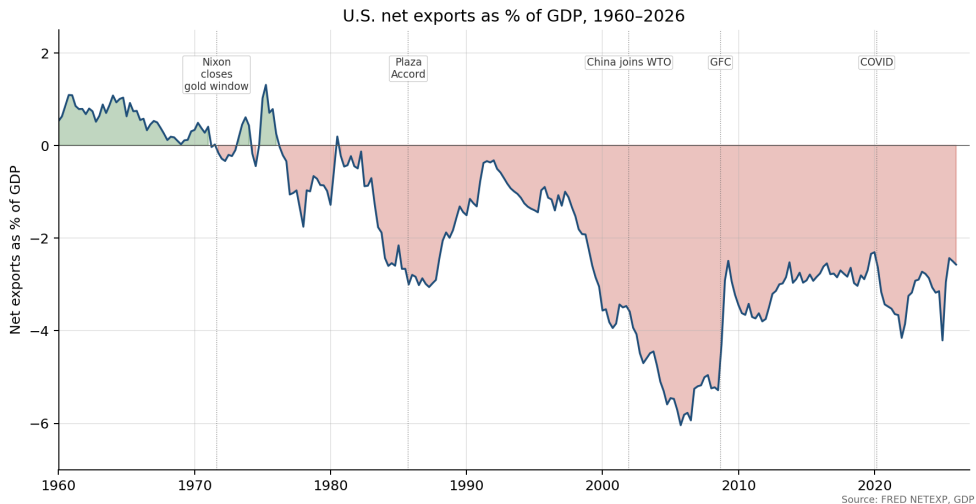
$NX > 0$, Size of the trade surplus = NX

► **Trade deficit:**

spending > output and imports > exports

$NX < 0$, Size of the trade deficit = $-NX$

U.S. net exports as % of GDP, 1960–2026



Source: FRED NETEXP / GDP. Positive = surplus (green), negative = deficit (red).

International capital flows

- ▶ **Net capital outflow**

- $= S - I$

- = net outflow of “loanable funds”

- = net purchases of foreign assets

- = the country's purchases of foreign assets minus foreign purchases of domestic assets

- ▶ When $S > I$, country is a net *lender*.

- ▶ When $S < I$, country is a net *borrower*.

The link between trade and capital flows (1 of 2)

$$NX = Y - (C + I + G)$$

implies

$$NX = (Y - C - G) - I = S - I$$

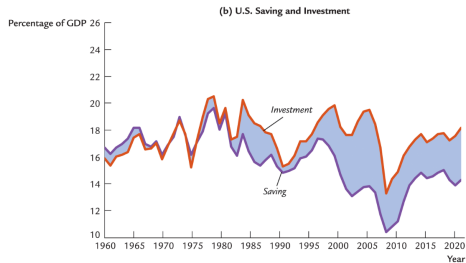
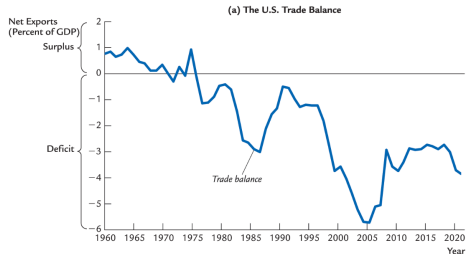
trade balance = net capital outflow

Thus,
a country with a trade deficit ($NX < 0$)
is a net borrower ($S < I$).

The link between trade and capital flows (2 of 2)

Trade Surplus	Balanced Trade	Trade Deficit
Exports $>$ Imports	Exports = Imports	Exports $<$ Imports
Net Exports $>$ 0	Net Exports = 0	Net Exports $<$ 0
$Y > C + I + G$	$Y = C + I + G$	$Y < C + I + G$
Saving $>$ Investment	Saving = Investment	Saving $<$ Investment
Net Capital Outflow $>$ 0	Net Capital Outflow = 0	Net Capital Outflow $<$ 0

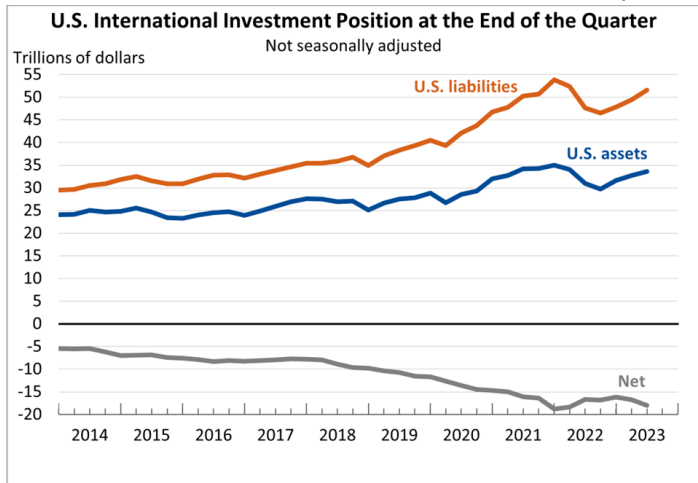
Saving, investment, and the trade balance 1960–2020



The United States: The world's largest debtor nation (1 of 2)

- ▶ Every year since the 1980s: huge trade deficits and net capital inflows (that is, net borrowing from abroad)
- ▶ As of July 2023:
 - ▶ U.S. residents owned \$33.58 trillion worth of foreign assets.
 - ▶ Foreigners owned \$51.58 trillion worth of U.S. assets.
 - ▶ U.S. net indebtedness to the rest of the world: \$18 trillion—higher than any other country, hence, the United States is the “world's largest debtor nation.”

The United States: The world's largest debtor nation (2 of 2)



Source: Bureau of Economic Analysis

Source: Bureau of Economic Analysis

Saving and investment in a small open economy (1 of 2)

- ▶ An open-economy version of the loanable funds model from Chapter 3
- ▶ Includes many of the same elements:

production function $Y = \bar{Y} = F(\bar{K}, \bar{L})$

consumption function $C = C(Y - T)$

investment function $I = I(r)$

exogenous policy variables $G = \bar{G}, T = \bar{T}$

Small open economy

- (a) Domestic and foreign bonds are perfect substitutes (same risk, maturity, etc.).
- (b) **Perfect capital mobility**: no restrictions on international trade in assets
- (c) Economy is **small**: cannot affect the world interest rate, denoted r^*

a & b imply $r = r^*$

c implies r^* is exogenous

Saving and investment in a small open economy (2 of 2)

$$NX = (Y - C - G) - I$$

$$NX = S - I$$

Assuming that $r = r^$*

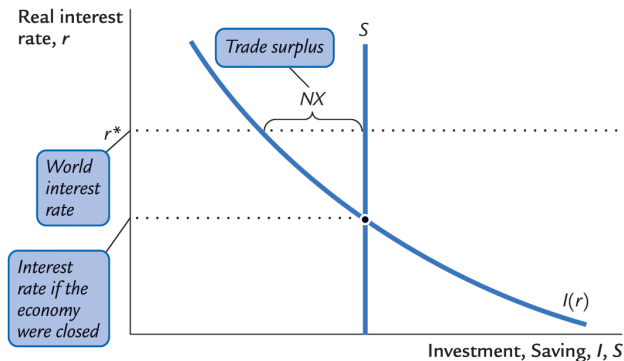
$$NX = (Y - C - G) - I = [\bar{Y} - C(\bar{Y} - T)] - I(r^*)$$

$$NX = \bar{S} - I(r^*)$$

But in a small open economy...

The exogenous world interest rate determines investment...

...and the difference between saving and investment determines net capital outflow and net exports.



Three experiments

1. Fiscal policy at home
2. Fiscal policy abroad
3. An increase in investment demand (exercise)

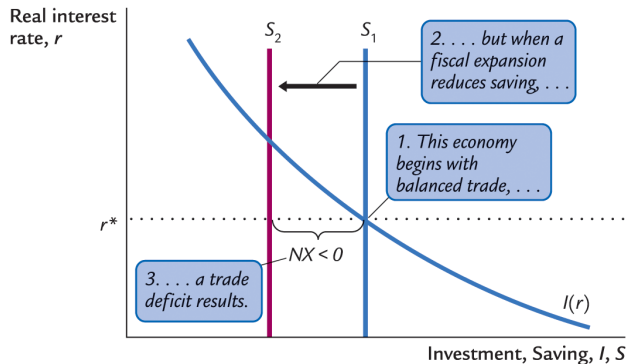
1. Fiscal policy at Home

An increase in G or decrease in T reduces saving.

Results:

$$\Delta I = 0$$

$$\Delta NX = \Delta S < 0$$



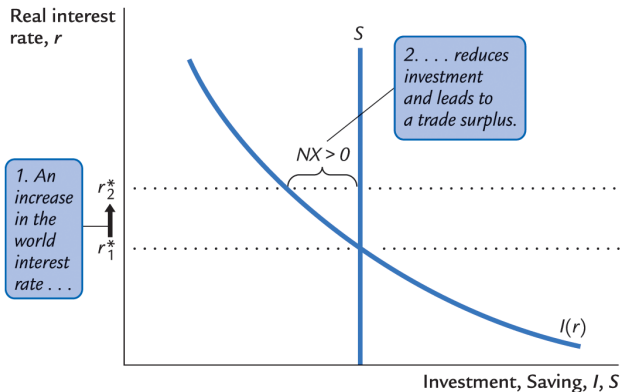
2. Fiscal policy abroad

Expansionary fiscal policy abroad raises the world interest rate.

Results:

$$\Delta I < 0$$

$$NX = -\Delta I > 0$$

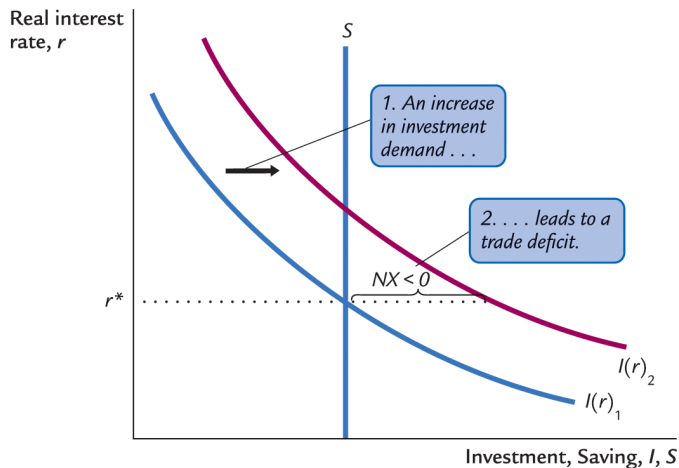


NOW YOU TRY: 3. An increase in investment demand

Use the model to determine the impact of an increase in investment demand on NX , S , I , and net capital outflow.

NOW YOU TRY: 3. An increase in investment demand, answers

$\Delta I > 0$,
 $\Delta S = 0$,
net capital outflow and
 NX fall by the amount ΔI



The nominal exchange rate

e = nominal exchange rate, the relative price of domestic currency in terms of foreign currency

(example: yen per dollar)

An increase in e implies that one U.S. dollar would buy more of Japanese Yen or any other currency.

An increase in e therefore means an **appreciation** of the dollar.

A few exchange rates, as of 12/15/2023

Country	Exchange rate
Euro area	0.91 euro/\$
Canada	1.33 C\$/
Japan	142.5 yen/\$
Mexico	16.97 pesos/\$
Russia	92.13 rubles/\$
South Africa	18.48 rand/\$
United Kingdom	0.79 pounds/\$

The real exchange rate (ε)

ε = real exchange rate

the relative price of domestic goods in terms of foreign goods
(example: Japanese Big Macs per U.S. Big Mac)

ε — the lowercase Greek letter *epsilon*

Understanding the units of ε

$$\varepsilon = \frac{e \times P}{P^*}$$

$$= \frac{(\text{Yen per \$}) \times (\$ \text{ per unit U.S. goods})}{\text{Yen per unit Japanese goods}}$$

$$= \frac{\text{Yen per unit U.S. goods}}{\text{Yen per unit Japanese goods}}$$

$$= \frac{\text{Units of Japanese goods}}{\text{per unit of U.S. goods}}$$

~ McZample ~

- ▶ One good: Big Mac
- ▶ Price in Japan:
 $P^* = 200$ yen
- ▶ Price in the United States:
 $P = \$2.50$
- ▶ Nominal exchange rate
 $e = 120$ yen/\$

$$\varepsilon = \frac{e \times P}{P^*} = \frac{120 \times 2.50}{200} = 1.5$$

To buy a U.S. Big Mac, someone from Japan would pay enough to buy 1.5 Japanese Big Macs.



Wikimedia,
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▶ Real data

ε in the real world and our model

- ▶ In the real world:
We can think of ε as the relative price of a basket of domestic goods in terms of a basket of foreign goods.
- ▶ In our macro model:
There's just one good, "output."
So ε is the relative price of one country's output in terms of the other country's output.

How NX depends on ε

If ε rises:

- ▶ U.S. goods become more expensive relative to foreign goods
- ▶ exports fall, imports rise
- ▶ net exports fall

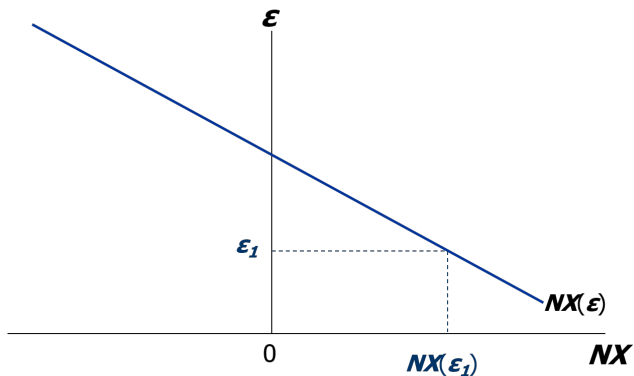
The net exports function

The **net exports function** reflects this inverse relationship between NX and ε :

$$NX = NX(\varepsilon).$$

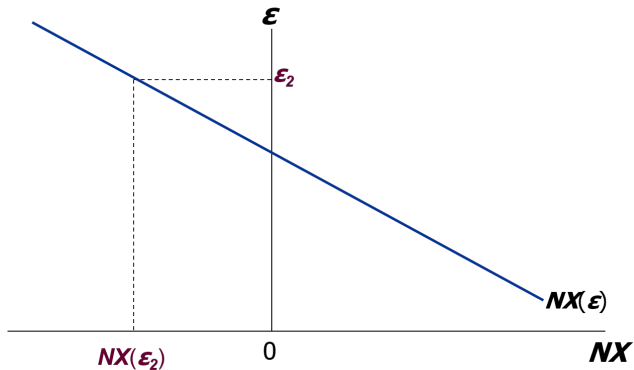
The NX curve for the United States, part 1

When ε is relatively low, U.S. goods are relatively inexpensive...
...so U.S. net exports will be high.



The NX curve for the United States, part 2

At high enough values of ε , U.S. goods become so expensive that we export less than we import.



How ε is determined (1 of 2)

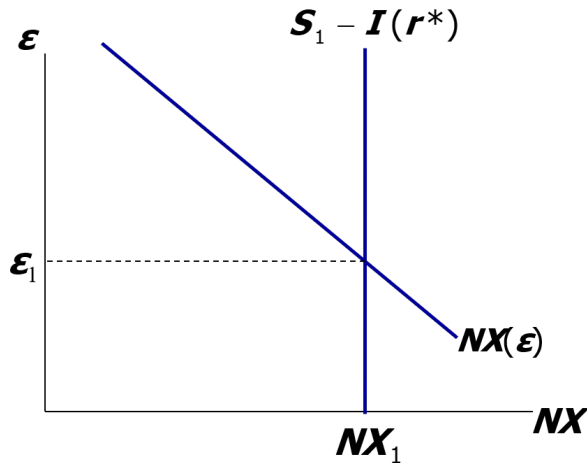
- ▶ The accounting identity says $NX = S - I$
- ▶ We saw earlier how $S - I$ is determined:
 - ▶ S depends on domestic factors (output, fiscal policy variables, etc.)
 - ▶ I is determined by the world interest rate r^*
- ▶ So, ε must adjust to ensure

$$NX(\varepsilon) = \bar{S} - I(r^*)$$

How ε is determined (2 of 2)

Neither S nor I depends on ε ,
so the net capital outflow
curve is vertical.

ε adjusts to equate NX with
net capital outflow, $S - I$.



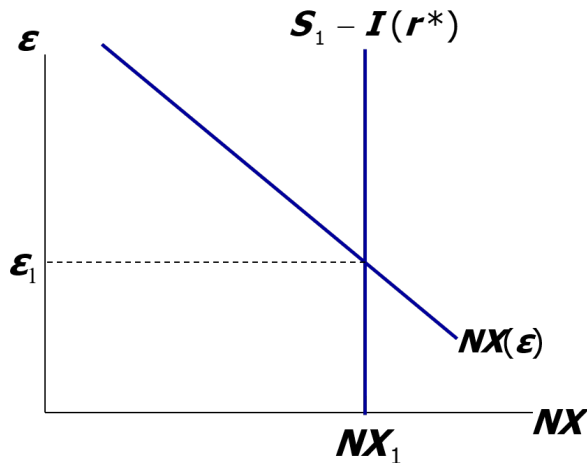
Interpretation: Supply and demand in the foreign exchange market

Demand:

Foreigners need dollars to buy U.S. net exports.

Supply:

Net capital outflow ($S - I$) is the supply of dollars to be invested abroad.



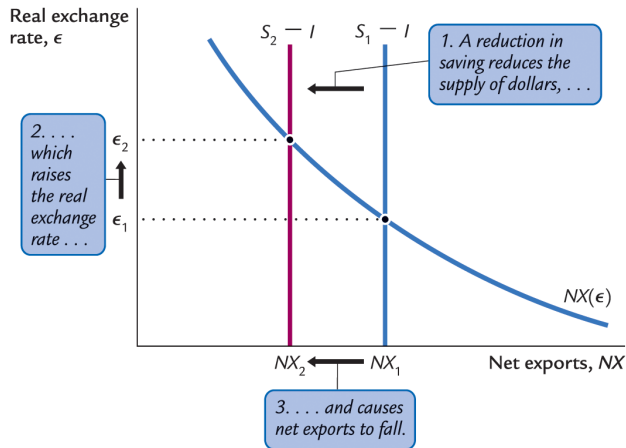
Four experiments

1. Fiscal policy at home
2. Fiscal policy abroad
3. An increase in investment demand (exercise)
4. Trade policy to restrict imports

1. Fiscal policy at Home

A fiscal expansion reduces national saving, net capital outflow, and the supply of dollars in the foreign exchange market...

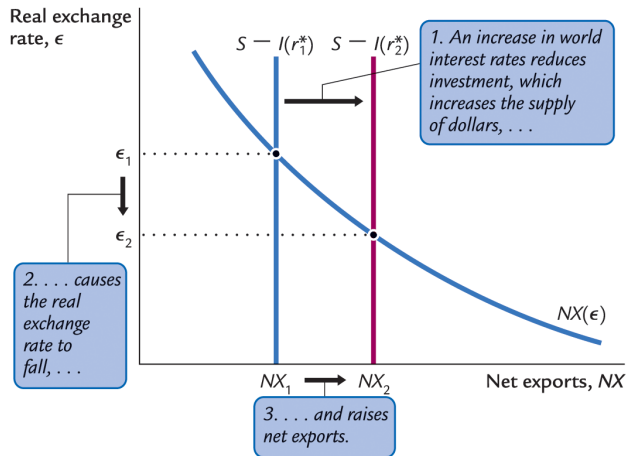
...causing the real exchange rate to rise and NX to fall.



2. Fiscal policy abroad

An increase in r^* reduces investment, increasing net capital outflow and the supply of dollars in the foreign exchange market...

...causing the real exchange rate to fall and NX to rise.



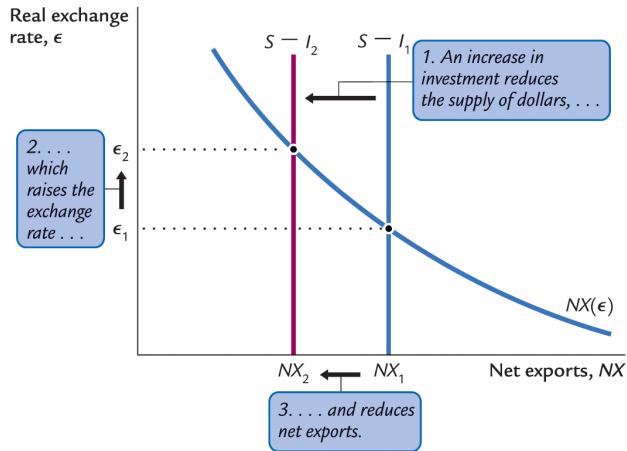
NOW YOU TRY: 3. Increase in investment demand

Determine the impact of an increase in investment demand on net exports, net capital outflow, and the real exchange rate.

NOW YOU TRY: 3. Increase in investment demand, answers

An increase in investment reduces net capital outflow and the supply of dollars in the foreign exchange market...

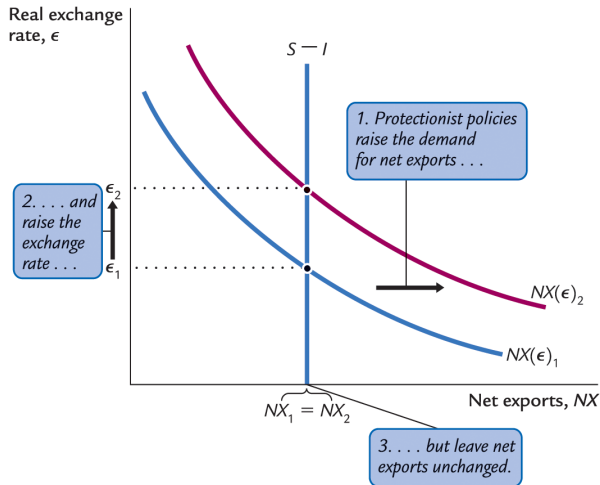
...causing the real exchange rate to rise and NX to fall.



4. Trade policy to restrict imports, part 1

At any given ϵ , an import tariff reduces IM , increases NX , and increases demand for dollars.

Trade policy doesn't affect S or I , so capital flows and the supply of dollars remain fixed.



4. Trade policy to restrict imports, part 2

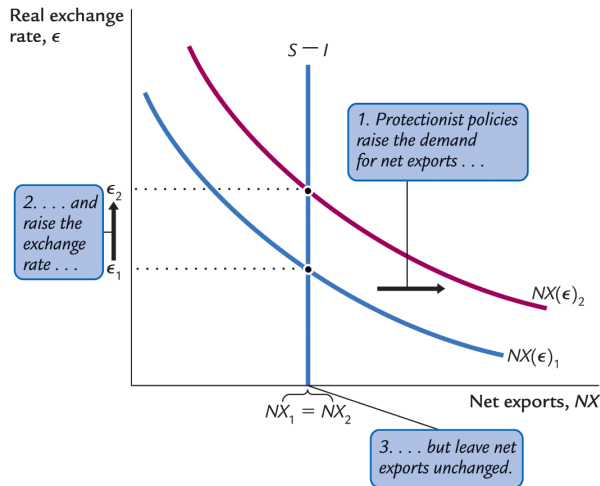
Results:

$\Delta \epsilon > 0$ (demand
increase)

$\Delta NX = 0$ (supply fixed)

$\Delta IM < 0$ (policy)

$\Delta EX < 0$ (rise in ϵ)



The determinants of the nominal exchange rate, part 1

- ▶ Start with the expression for the real exchange rate:

$$\varepsilon = \frac{e \times P}{P^*}$$

- ▶ Solve for the nominal exchange rate:

$$e = \varepsilon \times \frac{P^*}{P}$$

The determinants of the nominal exchange rate, part 2

So e depends on the real exchange rate and the price levels at home and abroad... and we know how each of them is determined:

$$e = \varepsilon \times \frac{P^*}{P}$$

$$NX(\varepsilon) = \bar{S} - I(r^*)$$

$$\frac{M}{P} = L(r^* + \pi, Y)$$

$$\frac{M^*}{P^*} = L^*(r^* + \pi^*, Y^*)$$

The determinants of the nominal exchange rate, part 3

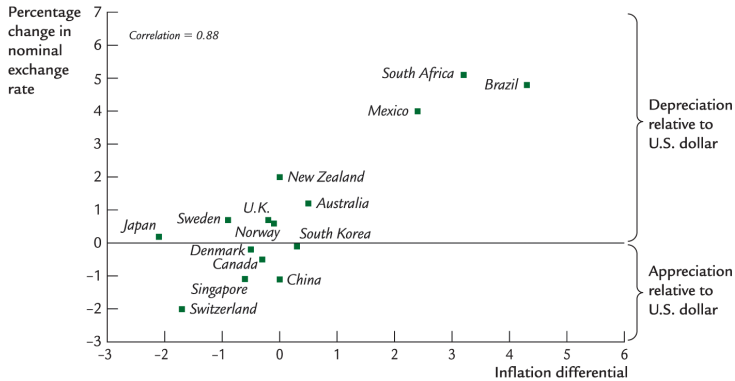
$$e = \varepsilon \times \frac{P^*}{P}$$

Rewrite this equation in growth rates:

$$\frac{\Delta e}{e} = \frac{\Delta \varepsilon}{\varepsilon} + \frac{\Delta P^*}{P^*} - \frac{\Delta P}{P} = \frac{\Delta \varepsilon}{\varepsilon} + \pi^* - \pi$$

For a given value of ε , the growth rate of e equals the difference between foreign and domestic inflation rates.

Inflation differentials and nominal exchange rates for a cross section of countries



Purchasing-power parity (PPP), part 1

Two definitions:

- ▶ a doctrine that states that goods must sell at the same (currency-adjusted) price in all countries
- ▶ the nominal exchange rate adjusts to equalize the cost of a basket of goods across countries

Reasoning: arbitrage, the law of one price

Purchasing-power parity (PPP), part 2

PPP: $e \times P = P^*$

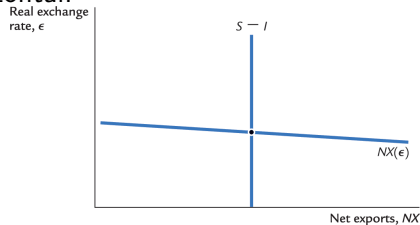
- ▶ Solve for e : $e = P^*/P$
- ▶ PPP implies that the nominal exchange rate between two countries equals the ratio of the countries' price levels.

Purchasing-power parity (PPP), part 3

If $e = P^*/P$, then

$$\varepsilon = e \times \frac{P}{P^*} = \frac{P^*}{P} \times \frac{P}{P^*} = 1$$

and the NX curve is horizontal:



Under PPP, changes in $(S - I)$ have no impact on ε or e .

Does PPP hold in the real world?

No, for two reasons:

1. International arbitrage is not possible
 - ▶ nontraded goods
 - ▶ transportation costs
2. Different countries' goods are not perfect substitutes

Yet PPP is a useful theory:

- ▶ It's simple and intuitive.
- ▶ In the real world, nominal exchange rates tend toward their PPP values over the long run.

The United States as a large open economy

- ▶ So far, we've learned long-run models for two extreme cases:
 - ▶ closed economy (Chapter 3)
 - ▶ small open economy (Chapter 6)
- ▶ A large open economy—like the United States—falls between these two extremes.
- ▶ The results from large open-economy analysis are a mixture of the results for the closed and small open-economy cases.

A fiscal expansion in three models

A fiscal expansion causes national saving to fall. The effects depend on openness and size.

	Closed Economy	Large Open Economy	Small Open Economy
r	Rises	Rises, but not as much	No change
I	Falls	Falls, but not as much	No change
NX	No change	Falls, but not as much	Falls

Chapter Summary, Part 1

- ▶ Net exports—the difference between:
 - ▶ exports and imports
 - ▶ a country's output (Y) and its spending ($C + I + G$)
- ▶ Net capital outflow equals:
 - ▶ purchases of foreign assets minus foreign purchases of the country's assets
 - ▶ the difference between saving and investment

Chapter Summary, Part 2

- ▶ National income accounts identities
 - ▶ $Y = C + I + G + NX$
 - ▶ trade balance $NX = S - I$ net capital outflow
- ▶ Impact of policies on NX
 - ▶ NX increases if policy causes S to rise or I to fall
 - ▶ NX does not change if policy affects neither S nor I (example: trade policy)

Chapter Summary, Part 3

- ▶ Exchange rates
 - ▶ nominal: the price of a country's currency in terms of another country's currency
 - ▶ real: the price of a country's goods in terms of another country's goods
- ▶ The real exchange rate equals the nominal rate times the ratio of prices of the two countries.

Chapter Summary, Part 4

- ▶ How the real exchange rate is determined:
 - ▶ NX depends negatively on the real exchange rate, other things equal.
 - ▶ The real exchange rate adjusts to equate NX with net capital outflow.

Chapter Summary, Part 5

- ▶ How the nominal exchange rate is determined:
 - ▶ e equals the real exchange rate times the country's price level relative to the foreign price level.
 - ▶ For a given value of the real exchange rate, the percentage change in the nominal exchange rate equals the difference between the foreign and domestic inflation rates.

Appendix: The Big Mac Index — PPP with real data (Jan 2024)

Concept: if PPP held, a Big Mac would cost the same everywhere in USD. Deviations tell you how over/under-valued a currency is versus the dollar.

Country	Local price	Local/USD ER	USD price	vs. US
United States	\$5.69	1.00	\$5.69	—
Switzerland	CHF 7.10	0.85	\$8.35	+47%
Norway	NOK 62	9.90	\$6.26	+10%
Euro area	€5.29	0.92	\$5.75	+1%
Japan	¥450	148	\$3.04	−47%
China	CNY 24	7.16	\$3.35	−41%
Turkey	TRY 129	29.7	\$4.34	−24%

Formula: $\text{USD price} = \frac{\text{Local price}}{\text{Local/USD ER}}$, then % deviation vs. US.

◀ back

